

# Randomized Algorithms

**Equalitarianism** between  
inputs!

Why randomized algorithms  
may be interesting?

No more **good** or **bad** inputs,  
only **good** or **bad** luck

# Randomized Algorithms

## Definition:

There are two kinds of randomized algorithms:

1. a *Las Vegas* algorithm returns a correct answer with probability 1, and we bound the expected number of steps it performs.
2. a *Monte Carlo* algorithm returns a correct answer with probability less than 1, and we bound the error probability and the worst case number of steps it performs.

# Randomized Algorithms

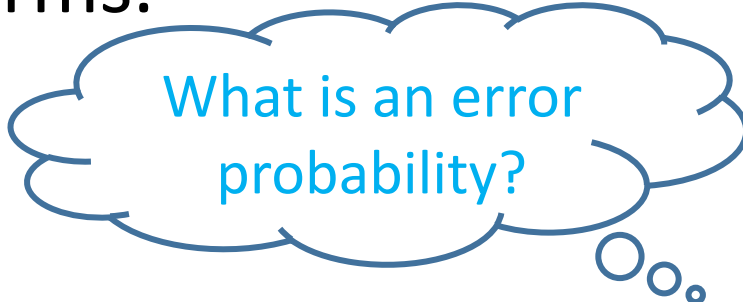
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2. a *Monte Carlo* algorithm returns a correct answer with probability less than 1, and we bound the error probability and the worst case number of steps it performs.

A blue thought bubble with a tail pointing towards the top right. Inside, the text "What is the expectation?" is written in blue. There are three small circles at the bottom right of the bubble.

What is the expectation?

A blue thought bubble with a tail pointing towards the bottom right. Inside, the text "What is an error probability?" is written in blue. There are three small circles at the bottom right of the bubble.

What is an error probability?

# Basic Probability

## Definitions:

1.  $\Omega$  is a **space** (set).
2.  $A \subseteq \Omega$  is an **event**.
3.  $P: \Omega \mapsto [0,1]$  is a **probability** function if  $\sum_{\omega \in \Omega} P(\omega) = 1$ .
4. A **probability space** is a pair  $(\Omega, P)$ .
5. A **random variable** is a function  $X: \Omega \mapsto \mathbb{R}$ .

We can define the sum of random variables as:

$$(X + Y)(\omega) = X(\omega) + Y(\omega).$$

# Basic Probability

## Example:

Tossing a coin:

$\Omega = \{0,1\}$  is the **space**

$A = \{0\} \subseteq \Omega$  is a possible **event**

$P: \Omega \mapsto [0,1]$  where  $P(0)=1/2$  and  $P(1)=1/2$  is a **probability** function. Note that  $\sum_{\omega \in \Omega} P(\omega) = 1$ .

Consider the following game:

If the coin toss result is 1, I get 1\$, otherwise I get 0\$.

$X$  can be a **random variable** representing my game profit.

# Basic Probability

## Definitions (cont.):

6. For a discrete variable  $X$ , the **expectation** of  $X$  is:

$$E[X] = \sum_{\omega \in \Omega} X(\omega) \cdot P(\omega)$$

In the last slide game example the expected profit value is:

$$E[X] = 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{1}{2}$$

The expectation is a linear operator:

- $E[X + Y] = E[X] + E[Y]$
- $E[c \cdot X] = c \cdot E[X]$

# Randomized QuickSort

## Randomized QuickSort Scheme:

Input: a set  $S$  of size  $n$ .

Output:  $S$  ordered from minimal to maximal element.

- 1) Choose *at random* an element  $y \in S$  as a pivot.
- 2) By comparison find the following sets:
$$S_1 = \{x \in S \mid x < y\},$$
$$S_2 = \{x \in S \mid x > y\}$$
- 3) Recursively sort  $S_1$  and  $S_2$
- 4) Return  $\langle \text{sorted } S_1, y, \text{sorted } S_2 \rangle$

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How to compute  
step 1?



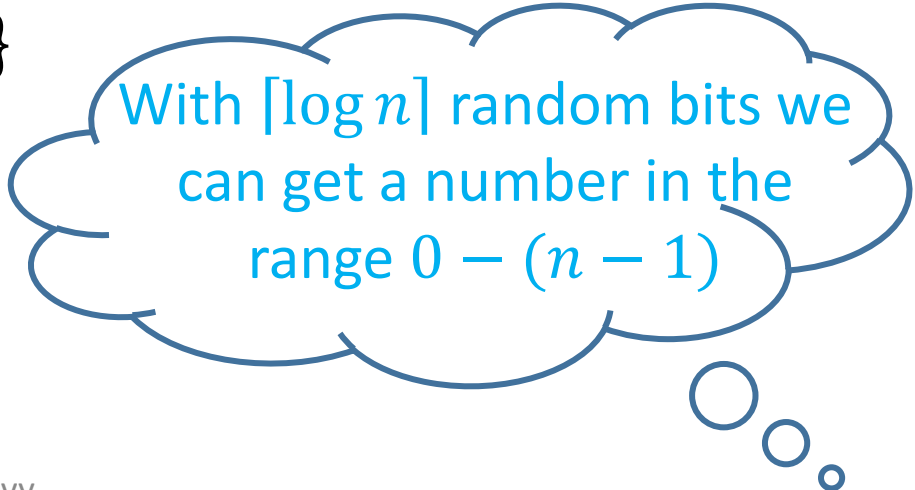
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With  $\lceil \log n \rceil$  random bits we  
can get a number in the  
range  $0 - (n - 1)$

# Randomized QuickSort

Example:

$S = \{7, 5, 2, 8, 9, 1, 4, 6\}, n=8$

Pos: 0 1 2 3 4 5 6 7

# Randomized QuickSort

Example:

$$S = \{7, 5, 2, 8, 9, 1, 4, 6\}, n=8$$

Pos: 0 1 2 3 4 5 6 7

Random bits: 100

$y=9$

$$S_1 = \{7, 5, 2, 8, 1, 4, 6\}$$

$$S_2 = \{ \}$$

# Randomized QuickSort

Example:

$$S = \{7, 5, 2, 8, 9, 1, 4, 6\}, n=8$$

Pos: 0 1 2 3 4 5 6 7

Random bits: 100

$y=9$

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Pos: 0 1 2 3 4 5 6

$$S_2 = \{\}$$

Random bits: 010

$y=2$

$$S_1 = \{1\}$$

$$S_2 = \{7, 5, 8, 4, 6\}$$

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Pos: 0 1 2 3 4 5 6

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Random bits: 010

$y=2$

$$S_1 = \{1\}$$

$$S_2 = \{7, 5, 8, 4, 6\}$$

Pos: 0 1 2 3 4

Random bits: 001

$y=5$

$$S_1 = \{4\}$$

$$S_2 = \{7, 8, 6\}$$

# Randomized QuickSort

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Pos: 0 1 2

Random bits: 00

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Pos: 0 1 2

$$S_2 = \{6, 7, 8\}$$

Random bits: 00

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Pos: 0 1 2 3 4 5 6 7

Random bits: 100

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Pos: 0 1 2 3 4 5 6

$$S_2 = \{\}$$

Random bits: 010

$y=2$

$$S_1 = \{1\}$$

$$S_2 = \{7, 5, 8, 4, 6\}$$

Pos: 0 1 2 3 4

$$S_2 = \{4, 5, 6, 7, 8\}$$

Random bits: 001

$y=5$

$$S_1 = \{4\}$$

$$S_2 = \{7, 8, 6\}$$

Pos: 0 1 2

$$S_2 = \{6, 7, 8\}$$

Random bits: 00

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$S = \{1, 2, 4, 5, 6, 7, 8, 9\}$

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$y=9$

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Random bits: 00

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# Randomized QuickSort

Complexity depends on  
the random bits

$= \{7, 5, 2, 8, 9, 1, 4, 6\}, n=8$

Pos: 0 1 2 3 4 5 6 7

$S = \{1, 2, 4, 5, 6, 7, 8, 9\}$

Random bits: 100

$y=9$

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# Randomized QuickSort

In fact,

The Complexity=Number of Comparisons

which is a random variable  $X$ !

How to compute the  
expectation of  $X$ ?

# Randomized QuickSort - Analysis

We will compute the expected number of comparisons that the algorithm performs for any input.

For  $1 \leq i \leq n$  denote by  $S^i$  the element in  $S$  that is in the  $i$ -th position in the sorted list.

**For example:** If  $S = \{7, 5, 2, 8, 9, 1, 4, 6\}$  then after the sorting we have:

$$S = \{1, 2, 4, 5, 6, 7, 8, 9\}$$

so,

$$S^1 = 1, S^2 = 2, S^3 = 4, S^4 = 5, S^5 = 6, S^6 = 7, S^7 = 8, S^8 = 9$$

# Randomized QuickSort - Analysis

We will look at a specific running of the algorithm and define the following set of random variables values according to the execution.

For  $1 \leq i, j \leq n$ ,

$$X_{i,j} = \begin{cases} 1, & \text{if } S^i \text{ is compared to } S^j \\ 0, & \text{otherwise} \end{cases}$$

# Randomized QuickSort - Analysis

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$$X_{1,2} = 1$$

$$X_{1,3} = 0$$

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$$X_{1,5} = 0$$

$$X_{1,6} = 0$$

$$X_{1,7} = 0$$

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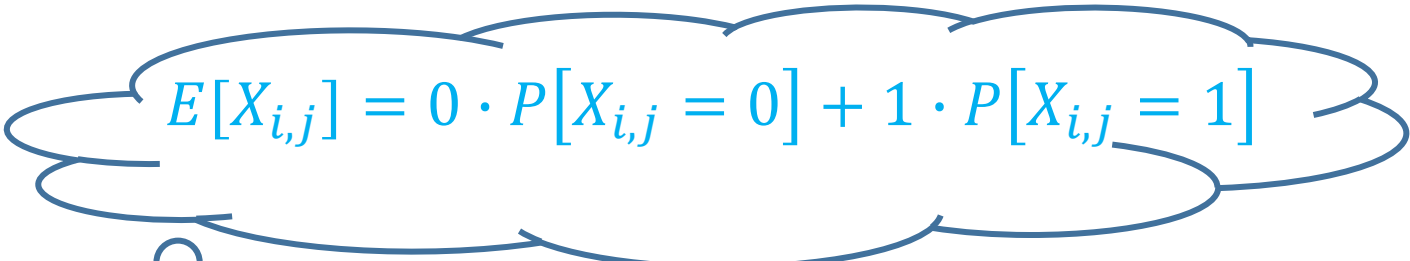
$$X_{6,8} = 1 \quad X_{5,8} = 1$$

$$X_{7,8} = 1$$

# Randomized QuickSort - Analysis

Analysis of expectation of comparisons number :

$$E[X] = E \left[ \sum_{i=1}^n \sum_{j>i} X_{ij} \right] = \sum_{i=1}^n \sum_{j>i} E[X_{ij}] = \sum_{i=1}^n \sum_{j>i} P[X_{ij} = 1]$$


$$E[X_{i,j}] = 0 \cdot P[X_{i,j} = 0] + 1 \cdot P[X_{i,j} = 1]$$


# Randomized QuickSort - Analysis

Observe the first time that a pivot  $y = S^k$  is chosen where  $i \leq k \leq j$

Claim 1:  $X_{ij} = 1$  if and only if  $k \in \{i, j\}$

Proof: Assume  $X_{ij} = 1$ .

Then,  $S^i$  is compared during run time with  $S^j$ . Note that until  $y = S^k$  is chosen where  $i \leq k \leq j$ ,  $S^i$  and  $S^j$  both belong to same set, either to  $S_1$  or to  $S_2$  and are only compared to the pivots but not to each other.

Now, if it holds that  $i < k < j$ , then at this phase  $S^i$  will belong to  $S_1$  and  $S^j$  will belong to  $S_2$  and from now on they will never be compared (because elements are only compared to pivots chosen from the set they belong to). Therefore, it must be that:  $k \in \{i, j\}$ .

# Randomized QuickSort - Analysis

Observe the first time that a pivot  $y = S^k$  is chosen where  $i \leq k \leq j$

Proof: (cont.)

Assume  $k \in \{i, j\}$ .

Again, note that until  $y = S^k$  is chosen where  $i \leq k \leq j$ ,  $S^i$  and  $S^j$  both belong to same set, either to  $S_1$  or to  $S_2$  and are only compared to the pivots but not to each other.

Now, if either  $y = S^i$  or  $y = S^j$  is the pivot, the other is an element in this set and, therefore, must be compared to the pivot. Thus, we have that:  $X_{ij} = 1$ .



# Randomized QuickSort - Analysis

Observe the first time that a pivot  $y = S^k$  is chosen where  $i \leq k \leq j$

Claim 2:  $p_{ij} = P[X_{ij} = 1] = \frac{2}{j-i+1}$

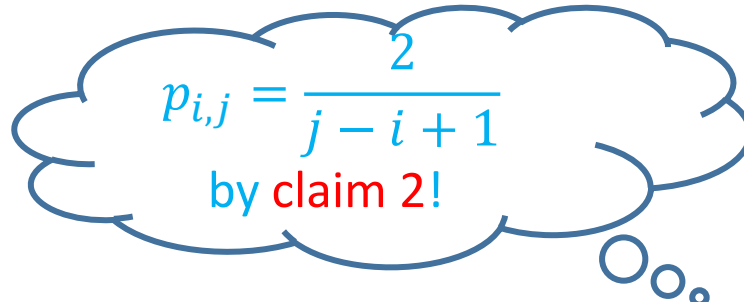
Proof:

There are only  $j - i + 1$  possible values that affect the probability that  $S^i$  and  $S^j$  are compared (because in the other cases they still belong to the same set), by claim 1.

By claim 1, the first time that a pivot  $y = S^k$  is chosen where  $i \leq k \leq j$  only two choices will cause that  $X_{ij} = 1$ .

# Randomized QuickSort - Analysis

Analysis of expectation of comparisons number :

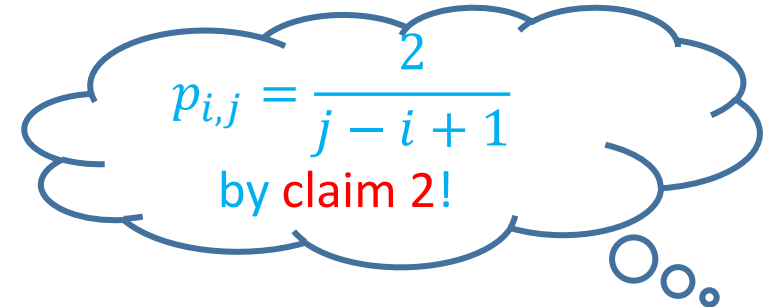
$$\begin{aligned} E[X] &= E \left[ \sum_{i=1}^n \sum_{j>i} X_{ij} \right] = \sum_{i=1}^n \sum_{j>i} E[X_{ij}] = \sum_{i=1}^n \sum_{j>i} P[X_{ij} = 1] \\ &= \sum_{i=1}^n \sum_{j>i} p_{ij} = \end{aligned}$$


$p_{i,j} = \frac{2}{j-i+1}$   
by claim 2!

# Randomized QuickSort - Analysis

Analysis of expectation of comparisons number :

$$\begin{aligned} E[X] &= E\left[\sum_{i=1}^n \sum_{j>i} X_{ij}\right] = \sum_{i=1}^n \sum_{j>i} E[X_{ij}] = \sum_{i=1}^n \sum_{j>i} P[X_{ij} = 1] \\ &= \sum_{i=1}^n \sum_{j>i} p_{ij} = \sum_{i=1}^n \sum_{j>i} \frac{2}{j-i+1} = \sum_{i=1}^n \sum_{k=1}^{n-i} \frac{2}{k+1} \end{aligned}$$


$$p_{i,j} = \frac{2}{j-i+1}$$

by claim 2!

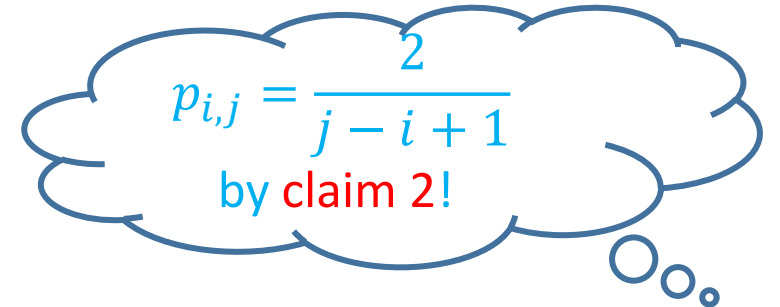
# Randomized QuickSort - Analysis

Analysis of expectation of comparisons number :

$$E[X] = E\left[\sum_{i=1}^n \sum_{j>i} X_{ij}\right] = \sum_{i=1}^n \sum_{j>i} E[X_{ij}] = \sum_{i=1}^n \sum_{j>i} P[X_{ij} = 1]$$

$$= \sum_{i=1}^n \sum_{j>i} p_{ij} = \sum_{i=1}^n \sum_{j>i} \frac{2}{j-i+1} = \sum_{i=1}^n \sum_{k=1}^{n-i} \frac{2}{k+1} \leq$$

$$2 \sum_{i=1}^n \sum_{k=1}^n \frac{1}{k+1}$$



$p_{i,j} = \frac{2}{j-i+1}$   
by claim 2!

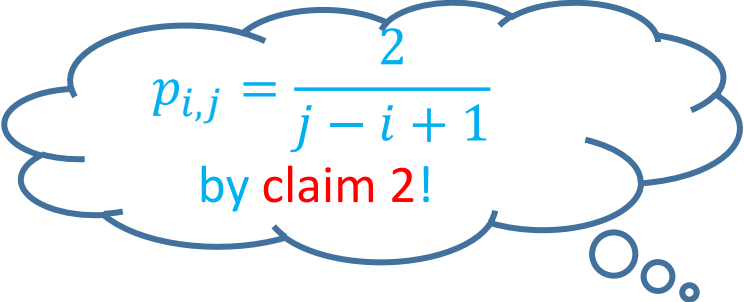
# Randomized QuickSort - Analysis

Analysis of expectation of comparisons number :

$$E[X] = E\left[\sum_{i=1}^n \sum_{j>i} X_{ij}\right] = \sum_{i=1}^n \sum_{j>i} E[X_{ij}] = \sum_{i=1}^n \sum_{j>i} P[X_{ij} = 1]$$

$$= \sum_{i=1}^n \sum_{j>i} p_{ij} = \sum_{i=1}^n \sum_{j>i} \frac{2}{j-i+1} = \sum_{i=1}^n \sum_{k=1}^{n-i} \frac{2}{k+1} \leq$$

$$2 \sum_{i=1}^n \sum_{k=1}^n \frac{1}{k+1} \leq 2 \sum_{i=1}^n \sum_{k=1}^n \frac{1}{k} =$$



$p_{i,j} = \frac{2}{j-i+1}$   
by claim 2!

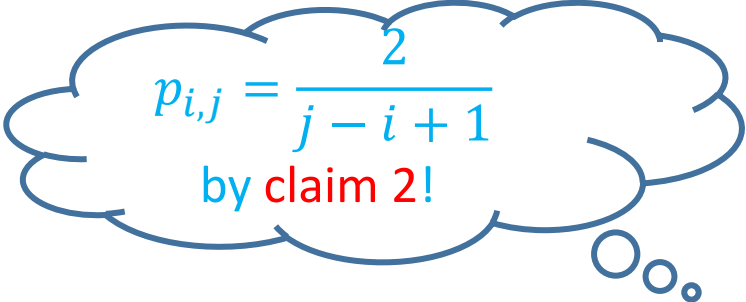
# Randomized QuickSort - Analysis

Analysis of expectation of comparisons number :

$$E[X] = E\left[\sum_{i=1}^n \sum_{j>i} X_{ij}\right] = \sum_{i=1}^n \sum_{j>i} E[X_{ij}] = \sum_{i=1}^n \sum_{j>i} P[X_{ij} = 1]$$

$$= \sum_{i=1}^n \sum_{j>i} p_{ij} = \sum_{i=1}^n \sum_{j>i} \frac{2}{j-i+1} = \sum_{i=1}^n \sum_{k=1}^{n-i} \frac{2}{k+1} \leq$$

$$2 \sum_{i=1}^n \sum_{k=1}^n \frac{1}{k+1} \leq 2 \sum_{i=1}^n \sum_{k=1}^n \frac{1}{k} = 2 \sum_{i=1}^n \ln(n) = O(n \log n)$$


$$p_{i,j} = \frac{2}{j-i+1}$$

by claim 2!